

Commitment Mathematics v5.3

Core Summary with Essential Equations

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Commitment Mathematics (CM) is a relational formalism in which the commitment is the sole primitive. Relations are primary; substances — particles, agents, objects — are derivative roles constituted by commitments. The framework is atemporal, substrate-independent, and validated by internal coherence. All constants are derived; there are no free parameters.

1. The Primitive

C = (L, H, P, CV₀)

- L (Liable Authority) — owes exactly CV₀. Perspective: $\hat{V}_l = -CV_0$ (1-dimensional, fixed).
- H (Holding Entity) — realises value across six dimensions.
- P — pledge specification: what is owed.
- CV₀ — base value. Frame-independent and invariant under all operations except bilateral renegotiation or ANL.

L and H are roles constituted by the commitment, not pre-existing agents who enter it. The asymmetry between $\hat{V}_l$ (scalar) and $\hat{V}_h$ (six-dimensional) is constitutive, not incidental.

2. Three Generators (fixed composition order)

The three Generators are applied in the sequence $\hat{V} \rightarrow \hat{F} \rightarrow \hat{C}$. The ordering is not interchangeable.

Generator	Symbol	Formula
Dual Valuation	$\hat{V}$	$\hat{V}_h(C) = +CV_0 \cdot V \cdot A \cdot (1+T) \quad \hat{V}_l(C) = -CV_0$
Framing	$\hat{F}$	$\hat{F}(CV) = CV \cdot \Phi, \quad \Phi = \sum W_k e^{i\theta_k}$
Constraint	$\hat{C}$	$\hat{C}(\psi) = \psi e^{(-kd)}, \quad k = \ln 2$

Commitment tensor (full composition):

$$^k(C_j) = CV_0(j) \otimes [V_j, A_j, T_j] \otimes \Phi_j \otimes e^{(-kd_j)}$$

Canonical realised value — with ternary gate $\max(0, \mathcal{J})$:

$$CV_realized = CV_0 \cdot V \cdot A \cdot (1+T) \cdot \|\Phi\| \cdot \max(0, \mathcal{J}) \cdot e^{(-kd)}$$

$\mathcal{J} \in \{+1, 0, -1\}$ is the TEST verdict. $\max(0, \mathcal{J})$ gates value to zero on INCONCLUSIVE or FALSE.

3. Anchor Space and Fabric Coordinates

Anchor space: $\mathcal{A} = [0,1]^3 = \{(V, A, T)\}$

- V — visibility (accessibility of pledge content to H)
- A — assurance (structural reliability of the pledge)
- T — transferability (dynamical mobility of the commitment)

Per-commitment Fabric coordinates

A_j (Defense — structural coherence)

$R_o \times T_r \times S \times R_e \times I$ (Memory — any factor = 0 $\rightarrow \mu_{local} = 0$)

T_j (Pulse — dynamical excitation)

INC on A simultaneously increments D_{local} ; INC on T simultaneously increments λ_{local} (double effect, not two separate operations).

Network Fabric coordinates (MONITOR aggregates)

$(1/|\mathcal{C}|) \cdot \sum_j \|\nabla_i CV_o(j)\|$

$|\{\text{phase-coherent pairs}\}| / |\mathcal{C}|^2$

$(1/|\mathcal{C}|) \cdot \sum_j T_j \cdot A_j$

Phase-coherent pair (C_i, C_j) : requires $\mu_{local}(C_i) > 0$, $\mu_{local}(C_j) > 0$, and $\mathcal{J}(F_i, F_j) = +1$.

Memory degradation (self-accelerating)

$\Delta\mu = -\ln 2 \cdot \sum_n [(\Delta D_n)^2 + (\Delta\lambda_n)^2 + (1-\mu_n)^2]$

The $(1-\mu)^2$ term produces positive feedback: degraded Memory accelerates further degradation. Memory re-storage must precede all other interventions.

4. Relational Distance and Interaction Modes

Relational distance (derived from frame phases):

$R_D(i,j) = (1/\pi) \cdot \arccos(\cos(\theta_i - \theta_j)) \in [0,1]$

Compensation efficiency (source of all mode thresholds):

$\text{Eff}(R_D) = e^{(-\ln 10 \cdot R_D)}$

Ternary intersection function $\mathcal{J}$ (TEST output)

- $\mathcal{J} = +1$ if $R_D \leq 0.2048$ ($\cos \Delta\theta \geq 0.8$) — frames agree
- $\mathcal{J} = 0$ if $0.2048 < R_D \leq 0.4030$ — NEGOTIATE required
- $\mathcal{J} = -1$ if $R_D > 0.4030$ — frames disagree

Interaction mode thresholds

- $R_D < e^{-3} \approx 0.050$ Coupling — all 11 operations; JMP possible
- $R_D < e^{-2} \approx 0.135$ Deep Interference — FLP, DRV available
- $R_D < e^{-1} \approx 0.368$ Interference — ROT, LNK available
- $R_D < 1 - e^{-1} \approx 0.632$ Exchange — mediated INC only; existing Channels persist

- $R_D \geq 1 - e^{-1}$ Isolation — no inter-commitment operations; all Channels dissolve

5. Eleven Base Operations

Hierarchy (higher levels override lower):

CRT / DRV / ANL > JMP / MRG / SPL > FLP / LNK > ROT > INC / OPN

Level 1 — Lifecycle

- CRT — create root commitment $C = (L, H, P, CV_0)$. No mode restriction.
- DRV — create/dissolve Derivative. Requires $R_D < e^{-2}$. $CV_0\text{der} = CV_0\text{root}$. Value decays 50% per depth ($e^{(-k d)}$, $k = \ln 2$). Cost on Sponsor only.
- ANL — annihilate commitment. Cascades immediately to all Derivatives (Root or Sponsor).

Level 2 — Topology-changing

- JMP — discrete state transition. Four conjunctive conditions: $R_D < e^{-3}$; $G_c = 1$ (gender complementarity); $\kappa > 0.8$; resistance barrier overcome. Irreversible.
- MRG — merge two compatible commitments. Coupling mode + Channel Integration trajectory ($\sigma_{ij} \geq \sigma_{\text{merge}}$).
- SPL — split one commitment into two. Governed by internal Channel trajectory ($\sigma_{\text{internal}} \leq \sigma_{\text{split}}$); mode-independent.

Level 3 — Discrete structural

- FLP — discrete symmetric inversion of one target $\in \{\text{role}, V, A, T\}$. Costless involution. $R_D < e^{-2}$ required.
- LNK — create / adjust / dissolve Channel σ_{ij} . $R_D < e^{-1}$ required. Bilateral consent.

Level 4 — Frame modification

- ROT — pure phase rotation $\theta_k \rightarrow \theta_k + \varphi \cdot R_k$; weights W_k unchanged. $R_D < e^{-1}$ required. Other-directed requires consent or pledge authorisation.

Level 5 — Minimal precondition

- INC — continuous parameter modification of V , A , or T (or compensation stock). Double effect: $\text{INC}_A \rightarrow D_{\text{local}}$; $\text{INC}_T \rightarrow \lambda_{\text{local}}$. L-resistance $R_L = \rho \cdot |\Delta CV|$, $\rho = 1 - (V \cdot A)$ on upward INC.
- OPN — Channel flow state operator. One discrete flow event. Triggered by EXECUTE ($\mathcal{F}=+1$) or VIOLATE ($\mathcal{F}=-1$). Zero cost.

Channel cost and throughput

$\sigma_{ij} \cdot T_i \cdot T_j$ (realised throughput)

$-(\ln 2) \cdot \Delta \sigma / 2$ per party (build cost; recovery = half on dissolution)

$+(\ln 2) \cdot \Delta \sigma / 2$ per party

Cycle stability (doom loops)

$\prod_e [\Phi_{ij} (1 + \sigma_{ij} \ln 2)] \cdot e^{(i \Delta \theta_e)}$ Stability $\Leftrightarrow |G_\gamma| < 1$ for every directed cycle γ

6. Coupling Coefficient and Gender Functions

$(1-A_j) \cdot \mu_{\text{local}}(C_j) \cdot (1-T_j)$ (carrier character: low A , high μ , low T)

$A_j \cdot (1-\mu_{\text{local}}(C_j)) \cdot T_j$ (receiver character: high A , low μ , high T)

1 iff $(M_A \approx 1 \text{ and } F_B \approx 1)$ or $(M_B \approx 1 \text{ and } F_A \approx 1)$

$[D_{\text{sys}} \cdot \mu_{\text{sys}} / ((1+\lambda_{\text{sys}})(1+R_D))] \cdot A(R_D)$ (JMP requires $\kappa > 0.8$)

where $A(R_D) = 1 - R_D^2$ if $|R_D| \leq 0.5$; else $2 - (R_D^2 - 1)$.

7. Two Relational Value Invariants

Both invariants use $m = 1 - |\sum e^{i\theta_j}| / N$ (global misalignment, $N = |E|$). $m = 0$ is perfect alignment; $m = 1$ is complete misalignment.

Stock invariant – RV_total (additive, path-invariant)

$\sum_j EV_j + \ln 2 \cdot (1-m) \cdot \sum_i <_j \Phi_{ij} + \ln 2 \cdot \sum_j \Psi_j - \ln 2 \cdot \sum_j (1-\mu_j) \Psi_j$

where:

- $EV_j = CV_j + \text{Comp}_j$ (Exchange Value: base value + compensation stock)
- $\Phi_{ij} = 1 - R_D(i,j)$ (pairwise relational closeness)
- $\Psi_j = G_j \cdot \mu_j \cdot D_j / (1+\lambda_j)$ (Generative Potential)
- $m = 1 - |\sum e^{i\theta_j}| / N$ (global misalignment)

The $(1-m)$ factor penalises misalignment dispersion. A network with high m contributes less coherence to RV_{total} regardless of mean proximity. The final negative term is the Memory Penalty – the only negative term m – and the only term that can grow autonomously through the $(1-\mu)^2$ self-feedback.

Flow invariant – RV_factor (multiplicative, per step)

$[(1+\Delta G)/(1+\Delta D)] \cdot [(B-m)/(B+m)] \cdot [D \cdot \mu / (1+\lambda)]$

where:

- $B = 1-e^{-1}$ (Isolation Boundary constant)
- ΔG = proportional change in Generative Output
- ΔD = proportional change in Derivation Output

$RV_{\text{factor}} = 1$ is stable flow. Any factor approaching zero produces structural collapse. $(B-m)/(B+m) \rightarrow 0$ as $m \rightarrow B$: the system reaches the Isolation Boundary and inter-commitment operations become impossible.

Alignment field equation (path-step discrete form)

$m(n) + \Delta^2 m(n) - \alpha \cdot \nabla^2_L m(n) = \kappa \cdot (1 - G_{\text{net}}(n))$

where:

- n – path traversal step
- $\Delta^2 m(n) = m(n+2) - 2m(n+1) + m(n)$ (path inertia)
- ∇^2_L – graph Laplacian over dependency graph (misalignment propagates through derivation edges)
- $G_{\text{net}} = G_S \cdot D_{\text{sys}} \cdot \mu_{\text{sys}} / (1+\lambda_{\text{sys}})$ (network generativity)

Zero misalignment ($m = 0$) requires $G_{\text{net}} = 1$. $G_{\text{net}} < 1$ drives m upward. The $\Delta^2 m$ term produces inertia: aligned states resist disruption; misaligned states self-reinforce. m and the alignment equation are distinct from

$m \varepsilon = \sigma(\{R_D\}) - \varepsilon$ is the statistical dispersion of relational distances used in earlier formulations; m is the current primary misalignment measure.

8. Fundamental Constants (all derived, no free parameters)

Constant / Threshold	Value	Role
k	$\ln 2$	Dependency decay, memory degradation, violation damage, Channel amplification
c	$\ln 10$	Compensation efficiency decay rate
τ_{high}	0.8	$\mathcal{J} = +1$ threshold ($\cos \Delta\theta \geq 0.8$)
τ_{low}	0.3	$\mathcal{J} = -1$ threshold ($\cos \Delta\theta < 0.3$)
μ_{critical}	$1/(2e) \approx 0.184$	MONITOR existence threshold
Coupling	$e^{-3} \approx 0.050$	All 11 operations; JMP possible
Deep Interference	$e^{-2} \approx 0.135$	FLP, DRV available
Exchange (onset)	$e^{-1} \approx 0.368$	ROT, LNK become unavailable
Isolation	$1 - e^{-1} \approx 0.632$	No inter-commitment operations; all Channels dissolve
ℓ_P	e^{-3}	Holographic Planck length (Coupling threshold)
B	$1 - e^{-1}$	Isolation Boundary — used in RV_factor

9. Generativity and Five-Dimensional Time

Generativity

$$G_S \cdot D \cdot \mu / (1 + \lambda) \quad (G_{vt} > 1 \text{ required for sustainable operation})$$

Maximum at Defensive Memory vertex ($D=1, \mu=1, \lambda=0$). λ in the denominator means Pulse reduces generative capacity. Unity vertex ($D=1, \mu=1, \lambda=1$) achieves only $G_{vt} = G_S/2$.

Five time dimensions (emergent from Fabric dynamics)

- T_1 — Operational: count of TEST-MONITOR cycles; driven by λ .
- T_2 — Mnemonic: $T_2 = -\ln \mu$; entropic accumulation from Memory degradation. Irreversible.
- T_3 — Relational: $T_3 = \Theta = (1/2\pi) \cdot \sum W_k \theta_k$; phase evolution through ROT. Reversible with cost.
- T_4 — Economic: $T_4 = \sum |Comp_j|$; cumulative compensation flow. Irreversible.
- T_5 — Generative: $T_5 = \int G_{vt} ds$ over topological operations. Irreversible.

Generative Temporal Anti-Correlation theorem

$$-D \cdot \mu / (1 + \lambda)^2 < 0 \quad (T_1 \text{ and } T_5 \text{ anti-correlated})$$

$$-D / (1 + \lambda) < 0 \quad (T_2 \text{ and } T_5 \text{ anti-correlated})$$

These are the only negative off-diagonal entries in the time metric tensor. Maximum generativity requires operational and mnemonic stillness.

10. Triple Control Architecture

Generative Control (episodic)

- TEST — coordinating primitive. Returns $\mathcal{J} \in \{+1, 0, -1\}$. Queries MONITOR for Fabric resources before every operation.
- NEGOTIATE — invoked on $\mathcal{J} = 0$. Incremental frame adjustment. Cost $C_{\text{neg}} = \gamma_{\text{neg}} \cdot CV_0$ per cycle.
- EXECUTE — invoked on $\mathcal{J} = +1$ AND MONITOR resource confirmation. Applies operation to k .
- VIOLATE — override on $\mathcal{J} = -1$. Cost $C_{\text{violate}} + \text{Damage} = k_v \cdot (\Delta D + \Delta \lambda + (1-\mu) \cdot |\Delta \mu|)$, $k_v = \ln 2$.
- OFFER — cooperative resolution on $\mathcal{J} = -1$ due to resource deficit. Requires Channel $\sigma_{ij} > 0$. Sequence: FLIP $\rightarrow$ OPEN $\rightarrow$ INC $\rightarrow$ re-TEST. Zero damage.

Governance Control (step-wise discrete — every TEST-MONITOR tick)

- MONITOR — computes Fabric coordinates; detects threshold breaches; signals HEAL. MONITOR exists only when $\mu_{\text{sys}} \geq \mu_{\text{critical}} = 1/(2e)$.
- SCRIBE — regulates μ_{sys} against spontaneous drift. Defends against DECEIVE.
- SECURE — regulates D against spontaneous drift. Defends against COERCE.
- MODULATE — regulates λ against runaway or depletion. Defends against EXTRACT.

Restorative Control (episodic — on failure or damage)

- HEAL — coordinating primitive. Receives signals from TEST ($\mathcal{J} = -1$) and MONITOR (threshold breach). Dispatches in order: HANDLE $\rightarrow$ REPAIR $\rightarrow$ IMPOSE.
- HANDLE — relational adjudication after control failure. Scope: individual commitment.
- REPAIR — restores Fabric coordinates D , μ , λ of an individual commitment (via INC, ROT, JMP). Not a base operation.
- IMPOSE — network-level intervention (forced CRT, ANL, mode assignment, Channel capacity reduction). Last resort. Requires HANDLE and REPAIR to be insufficient.

Repair sequencing rule (enforced by HEAL dispatch logic): Memory (μ) restoration precedes Defense (D) restoration precedes Pulse (λ) restoration. No exceptions.

11. Infrastructure Commitments

Infrastructure Commitments are commitments whose pledge P is to sustain one or more Fabric coordinates. They are governed by the same eleven base operations as all commitments. No special mechanisms exist.

- Reserve — absorbs and releases λ . Operated by MODULATE. Default: fully cloaked.
- Archive — preserves and restores μ . Operated by SCRIBE. Prerequisite to all other infrastructure. Default: fully cloaked.
- Boundary — reinforces D in a designated region. Operated by SECURE. Default: perimeter visible, internals cloaked.
- Sanctuary — protects a nascent commitment during development. Operated by HEAL. L is always the creating commitment. The Channel between L and the nascent commitment is constitutive of the Sanctuary (established at CRT, not by subsequent LINK).

- Firewall — maintains the Sanctuary perimeter; evaluates incoming commitments via TEST; executes enforcement action E on unauthorised commitments. Constitutively paired with every Sanctuary; neither functions without the other.

Infrastructure Access Condition

C_i can draw on $C_{\text{inf}} \iff$ Channel $\sigma_{i,\text{inf}} > 0$ exists

| All Channels dissolve on Isolation. Infrastructure in Isolation cannot perform its pledge.

12. Holographic Bound and Four Structural Streams

Holographic bound

$A_{\text{boundary}} / \ell_P^2, \ell_P = e^{-3}$

| Information capacity scales with boundary area, not volume or hierarchical depth. Organisational intelligence resides at interfaces.

Network capacity (sustainable generativity)

$G \cdot \mu \cdot D / (1 + \lambda)$

Four structural streams (each anchor parameter traces one complete path)

- Stream 1 $V \rightarrow \text{Frame } (\Phi) \rightarrow \text{Memory } (\mu) \rightarrow \text{SCRIBE} \rightarrow \text{Archive} \rightarrow \text{Temporal Entanglement} \rightarrow \text{Coherence: } \ln 2 \cdot (1 - m) \cdot \sum \Phi_{ij}$
- Stream 2 $A \rightarrow \text{Dependency depth } (d) \rightarrow \text{Defense } (D) \rightarrow \text{SECURE} \rightarrow \text{Boundary} \rightarrow \text{Fabric geometry} \rightarrow \text{Memory Penalty: } -\ln 2 \cdot \sum (1 - \mu_j) \Psi_j$
- Stream 3 $T \rightarrow \text{Base Value } (CV_0) \rightarrow \text{Pulse } (\lambda) \rightarrow \text{MODULATE} \rightarrow \text{Reserve} \rightarrow \text{Channels} \rightarrow \text{Exchange Value: } \sum EV_j$
- Stream 4 $CV_{\text{realized}} \rightarrow \text{Generative Potential } (\Psi) \rightarrow D + \mu + \lambda \rightarrow \text{HEAL} \rightarrow \text{Sanctuary+Firewall} \rightarrow \text{Generativity } (G_{\text{vt}}) \rightarrow RV_{\text{total}}$ conservation itself

| Stream 4 is the only stream that requires all three Fabric coordinates simultaneously. Its terminus is not one term of RV_{total} but the conservation of RV_{total} as a whole. The coefficient $\ln 2$ appears in all four stream termini — the formal signature of a single relational decay principle operating at every structural level.

Closure

Commitment Mathematics v5.3 is a complete, closed, atemporal relational formalism. Every structure is entailed by the sole primitive $C = (L, H, P, CV_0)$. The framework contains no free parameters. The constants $k = \ln 2$, $c = \ln 10$, and the mode thresholds e^{-3} , e^{-2} , e^{-1} , $1 - e^{-1}$ all follow from the structure of the compensation efficiency function and the conservation requirement. The three Generators, the eleven base operations, the triple control architecture, and the five Infrastructure Commitment types are each necessary; none is appended by design.